

Question ID: 0bd33265

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

The equation $h = \frac{9(v-273.15)}{5} + 32$ gives the corresponding temperature h , in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. If a substance has a temperature of **467.33** degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

Correct Answer: 515

Rationale

The correct answer is **515**. It's given that the equation $h = \frac{9(v-273.15)}{5} + 32$ gives the corresponding temperature h , in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. Substituting **467.33** for h in the given equation yields **467.33** $= \frac{9(v-273.15)}{5} + 32$. Subtracting **32** from both sides of this equation yields **435.33** $= \frac{9(v-273.15)}{5}$. Multiplying both sides of this equation by **5** yields **2,176.65** $= 9(v - 273.15)$. Dividing both sides of this equation by **9** yields **241.85** $= v - 273.15$. Adding **273.15** to both sides of this equation yields **515** $= v$. Therefore, if a substance has a temperature of **467.33** degrees Fahrenheit, the corresponding temperature, in kelvins, of this substance is **515**.